

# Code-Layout, Readability and Reusability

## 5. Project Development Phase | Credit Card Approval Prediction

| Prepared For                    | Prepared By  | Domain  | Date         |
|---------------------------------|--------------|---------|--------------|
| Credit Card Approval Prediction | Ankitha Vana | Finance | July 1, 2026 |

### Recommended Code Layout

```
credit-card-approval-prediction/  
  data/  
    application_record.csv  
    credit_record.csv  
  notebooks/  
    model_training.ipynb  
  models/  
    credit_card_model.pkl  
    preprocessing_pipeline.pkl  
  app.py  
  templates/  
    index.html  
    result.html  
  static/  
    style.css  
  requirements.txt  
  README.md
```

### Readability Practices

- Use clear function names such as load\_data, clean\_data, engineer\_features, train\_models, and predict\_approval.
- Keep preprocessing logic reusable between training and prediction.
- Store configuration values such as file paths and model names in one place.
- Add comments only where feature engineering or label conversion logic may be hard to understand.
- Separate notebook experimentation from production Flask code.

### Reusable Modules

| Module            | Responsibility                               |
|-------------------|----------------------------------------------|
| data_loader.py    | Read application and credit record files.    |
| preprocessing.py  | Clean, merge, encode, and transform data.    |
| model_training.py | Train and compare classification models.     |
| predict.py        | Load model and generate approval prediction. |
| app.py            | Expose the web routes and UI integration.    |